

ATHARV PATOLE

AI Engineer | Applied AI Engineer | Forward-Deployed Engineer

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SUMMARY

Four years shipping data and AI systems end to end: pipelines, retrieval, agents, evals, and the interface people actually open. I start with the people stuck on the problem, find what they need rather than what they ask for, and gate every release on evals.

WORK EXPERIENCE

Fulton Ring | Data Scientist, AI Systems | New York, NY

Aug 2025 - Jul 2026

- Housing nonprofit analysts queued behind engineers for every data pull. Shipped a production LLM search layer over housing, demographic, and environmental data so they could ask in plain language, cutting manual review ~40%.
- Retrieval returned plausible but wrong parcels and groundedness review took hours. Rebuilt Pinecone RAG on structured metadata and gated releases on LangSmith and LLM-as-judge evals, raising retrieval consistency ~30% and shrinking review to minutes.
- A federal catalog of infrastructure and resilience data needed a durable public home. Co-built the AI workflow pipeline, alongside two open-data partner organizations, that republished 400+ layers as cloud-native GeoParquet and PMTiles.
- Architects, engineers, and investors had no route into the property data without borrowing a data team. Launched standalone Next.js and MapLibre sites serving 10GB+ spatial files, trimming query latency ~60%.

Community Dream Foundation | Data Analyst | Remote, US

Aug 2025 - Jul 2026

- Reporting numbers drifted from source systems and reached stakeholders unchecked. Traced each mismatch back through the pipeline, then wrote Python and SQL validation checks that catch that class of error before every release.
- Internal tools and REST APIs quietly disagreed, and non-technical staff could not sanity-check the numbers. Verified data against source, stood up dashboards they read directly, and tracked every test scenario to closure with developers.

Scon Infra | Analytics Engineer | Mumbai, India

Jul 2020 - Jun 2023

- Finance and engineering leads reconciled infrastructure planning figures by hand each month, stalling sign-off. Scripted the reporting logic across \$12M+ in planning workflows, reducing recurring manual effort ~35%.
- Downstream models kept breaking on inconsistent source data. Engineered Python and SQL ingestion pipelines for real estate and construction data, with validation rules set alongside engineering, retiring manual prep entirely.
- Executives waited days on hand-assembled reports before committing. Defined governance rules and relational models spanning engineering and finance, shortening decision turnaround ~45%.

PROJECTS

SpiderNet | Natural-Language Property Intelligence for NYC | [App Link](#)

- Designed an agent that turns plain-English questions about any NYC lot into tool calls on a hand-written MCP server over parcel records, zoning and building code text, and flood risk, returning grounded answers mapped across ~30 layers.
- Deployed DuckDB inside Vercel serverless functions reading GeoParquet on Cloudflare R2 through HTTP range requests, and opened the same data through a free public REST API with an OpenAPI spec.
- A Kriging and XGBoost flood model extrapolated badly because sensors aren't randomly sited. Replaced it with physics-based HAND-stage inundation across 241 sensors, scoring ~1M NYC properties, validated on 311 complaints and DEP's Stormwater Map.

TECHNICAL SKILLS

LLMs and Agents: OpenAI, Anthropic Claude, and Gemini APIs, LLM agents, multi-step workflows, tool and function calling, MCP,

LangGraph, LangChain, OpenAI Agents SDK, Vercel AI SDK, structured outputs, prompt and context engineering, human-in-the-loop

Retrieval and Eval: Pinecone, pgvector, Chroma, LlamaIndex, hybrid retrieval (BM25 and dense), reranking, Graph RAG, LangSmith, Langfuse, Braintrust, OpenTelemetry, LLM-as-judge, regression evals, guardrails, prompt-injection defense, PII filtering

Machine Learning: PyTorch, Hugging Face, fine-tuning (LoRA), scikit-learn, XGBoost, classification, forecasting, anomaly detection

Backend and Frontend: Python, TypeScript, JavaScript, SQL, FastAPI, Flask, Node.js, REST APIs, webhooks, Next.js, React

Data, Cloud, Geospatial: AWS, GCP, Vercel, Cloudflare R2, Docker, Kubernetes, CI/CD (GitHub Actions), ETL/ELT, PySpark, Databricks, dbt, PostgreSQL, DuckDB, BigQuery, Snowflake, MongoDB, GDAL, GeoPandas, MapLibre GL, PMTiles, GeoParquet

Forward-Deployed and GTM: Customer discovery, Requirements scoping, Solution architecture, POCs and demos, Stakeholder communication, Technical documentation, Workflow automation (Clay), CRM integrations

EDUCATION AND AWARDS

Pace University | M.S. Computer Science | New York, NY

May 2025

Sanjay Ghodawat University | B.Tech Civil Engineering | India

May 2021

1st Place, Global Governance & Climate Hackathon: satellite-data tool attributing cross-border Amazon fires to their source

1st Place, FOAK Hackathon: capital stack and monetization model for a CO2-to-graphite startup's first commercial facility